

ELE228 - Programming for Circuits and Systems Preliminary Work 5

1 Questions

1. 1.1. Consider the circuit in *Fig. 1*. Derive the numerical expression of the transfer function $H(s) = \frac{V_{out}(s)}{V_{in}(s)}$ for $R_1 = 100 \text{ k}\Omega$ and $R_1 = 10 \text{ k}\Omega$. Then find $H(j\omega)$ and cut-off frequencies (f_c). Comment on the filtering characteristics of the input-output relationship.
- 1.2. Using MATLAB, write a code that finds the cut-off frequencies (f_c) from $H(s)$.
- 1.3. Using MATLAB, plot the Bode diagram of systems in logarithmic scale (for 2 different R_1 values in the same graph. Show cut-off frequencies in terms of Hz in the graph.

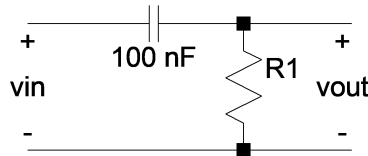


Figure 1

2. 2.1. Consider the circuit in *Fig. 2*. Derive the numerical expression of the transfer function $H(s) = \frac{V_{out}(s)}{V_{in}(s)}$ for $R_1 = 100 \Omega$ and $R_1 = 1 \text{ k}\Omega$. Then find $H(j\omega)$ and cut-off frequencies (f_c). Comment on the filtering characteristics of the input-output relationship.
- 2.2. Using MATLAB, write a code that finds the cut-off frequencies (f_c) from $H(s)$.
- 2.3. Using MATLAB, plot the Bode diagram of systems in logarithmic scale (for 2 different R_1 values in the same graph. Show cut-off frequencies in terms of Hz in the graph.

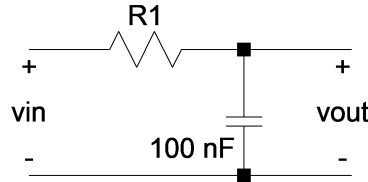


Figure 2

3. 3.1. Consider the circuit in *Fig. 3*. Derive the numerical expression of the transfer function $H(s) = \frac{V_{out}(s)}{V_{in}(s)}$ for $R_1 = R_2 = 10 \text{ k}\Omega$ and $R_3 = R_4 = 100 \Omega$. Then find $H(j\omega)$ and cut-off frequencies (f_c) and bandwidth. Comment on the filtering characteristics of the input-output relationship.
- 3.2. Using MATLAB, write a code that finds the cut-off frequencies (f_c) from $H(s)$.
- 3.3. Using MATLAB, plot the Bode diagram of systems for linear scale and logarithmic scale (for 2 different resistance values in the same graph). Show cut-off frequencies in terms of Hz in the graph. Compare linear and logarithmic scales and explain why we use the logarithmic scale in Bode. Explain the advantages and disadvantages of higher-order filters.

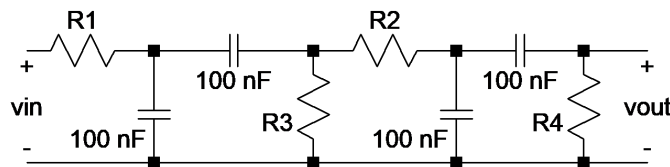


Figure 3

2 Answers

1. 1.1. Impedance of the capacitor in the s -domain: $\frac{1}{sC} = \frac{10^7}{s}$. By the voltage divider,

$$V_{\text{out}} = V_{\text{in}} \cdot \frac{R_1}{R_1 + \frac{10^7}{s}}, \quad \omega_c = \frac{1}{RC}, \quad f_c = \frac{\omega}{2\pi}$$

$$R_1 = 100 \text{ k}\Omega \implies H(s) = \frac{s}{s + 100}, \quad H(j\omega) = \frac{j\omega}{j\omega + 100}, \quad \omega_c = 100 \frac{\text{rad}}{\text{s}}, \quad f_c \approx 15.92 \text{ Hz}$$

$$R_1 = 10 \text{ k}\Omega \implies H(s) = \frac{s}{s + 1000}, \quad H(j\omega) = \frac{j\omega}{j\omega + 1000}, \quad \omega_c = 1000 \frac{\text{rad}}{\text{s}}, \quad f_c \approx 159.2 \text{ Hz}$$

Both are high-pass filters. As the resistance increases, the circuit passes the input voltage to the output voltage starting from lower frequencies. As the frequency of the input source decreases, the impedance of the capacitor increases. Therefore, the frequencies whose values are smaller than the cutoff value get attenuated.

- 1.2. Code:

```
H_100k = tf([1, 0], [1, 100]);
H_10k = tf([1, 0], [1, 1000]);

p_100k = pole(H_100k);
fc_100k = abs(p_100k) / (2*pi)

p_10k = pole(H_10k);
fc_10k = abs(p_10k) / (2*pi)
```

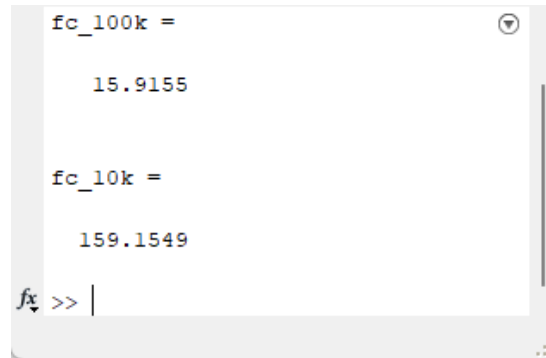


Figure 4

- 1.3. Code:

```
bode_opt = bodeoptions;
bode_opt.Grid = 'on';
bode_opt.FreqScale = 'log';
bode_opt.FreqUnits = 'Hz';
bode_opt.Title.String = 'Bode Plot of H(s)';
bode_opt.Title.FontWeight = 'bold';
bode_opt.Title.FontSize = 12;

figure;
bodeplot(H_100k, H_10k, bode_opt);
```

```

axes = findall(gcf,'type','axes');
for k = 1:length(axes)
    xline(axes(k),fc_100k,'--k', sprintf('f_{c,100k} = %.2f Hz',fc_100k));
    xline(axes(k),fc_10k,'--k', sprintf('f_{c,10k} = %.2f Hz',fc_10k));
end
legend('R_1 = 100 k\Omega','R_1 = 10 k\Omega','location','southwest');

```

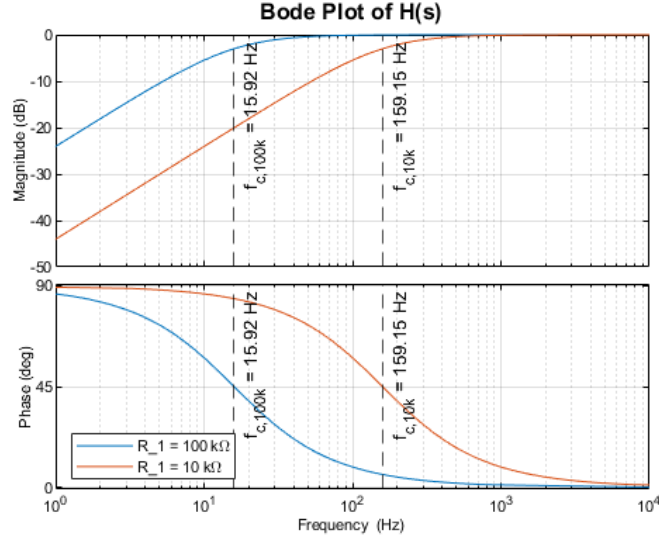


Figure 5

2. 2.1. Impedance of the capacitor in the s -domain: $\frac{1}{sC} = \frac{10^7}{s}$. By the voltage divider,

$$V_{\text{out}} = V_{\text{in}} \cdot \frac{\frac{10^7}{s}}{R_1 + \frac{10^7}{s}}, \quad \omega_c = \frac{1}{RC}, \quad f_c = \frac{\omega}{2\pi}$$

$$R_1 = 100 \, \Omega \implies H(s) = \frac{10^5}{s + 10^5}, \quad H(j\omega) = \frac{10^5}{j\omega + 10^5}, \quad \omega_c = 10^5 \text{ rad/s}, \quad f_c \approx 15915 \text{ Hz}$$

$$R_1 = 1 \text{ k}\Omega \implies H(s) = \frac{s}{s + 10^4}, \quad H(j\omega) = \frac{10^4}{j\omega + 10^4}, \quad \omega_c = 10^4 \text{ rad/s}, \quad f_c \approx 1591.5 \text{ Hz}$$

Both are low-pass filters. As the resistance increases, the circuit passes the input voltage to the resistor voltage starting from lower frequencies. As the frequency of the input source increases, the impedance of the capacitor decreases. Therefore, the frequencies whose values are greater than the cutoff value get attenuated.

2.2. Code:

```

H_100 = tf([0, 10^5], [1, 10^5]);
H_1000 = tf([0, 10^4], [1, 10^4]);

p_100 = pole(H_100);
fc_100 = abs(p_100) / (2*pi)

p_1000 = pole(H_1000);
fc_1000 = abs(p_1000) / (2*pi)

```

```
fc_100 =  
1.5915e+04  
  
fc_1000 =  
1.5915e+03  
fx >> |
```

Figure 6

2.3. Code:

```
bode_opt = bodeoptions;  
bode_opt.Grid = 'on';  
bode_opt.FreqScale = 'log';  
bode_opt.FreqUnits = 'Hz';  
bode_opt.Title.String = 'Bode Plot of H(s)';  
bode_opt.Title.FontWeight = 'bold';  
bode_opt.Title.FontSize = 11;  
  
figure;  
bodeplot(H_100, H_1000, bode_opt);  
  
axes = findall(gcf,'type','axes');  
for k = 1:length(axes)  
    xline(axes(k),fc_100,'--k',sprintf('f_{c,100} = %.2f Hz',fc_100));  
    xline(axes(k),fc_1000,'--k',sprintf('f_{c,1000} = %.2f Hz',fc_1000));  
end  
  
legend('R_1 = 100 \Omega', 'R_1 = 1000 \Omega', 'location', 'southwest');
```

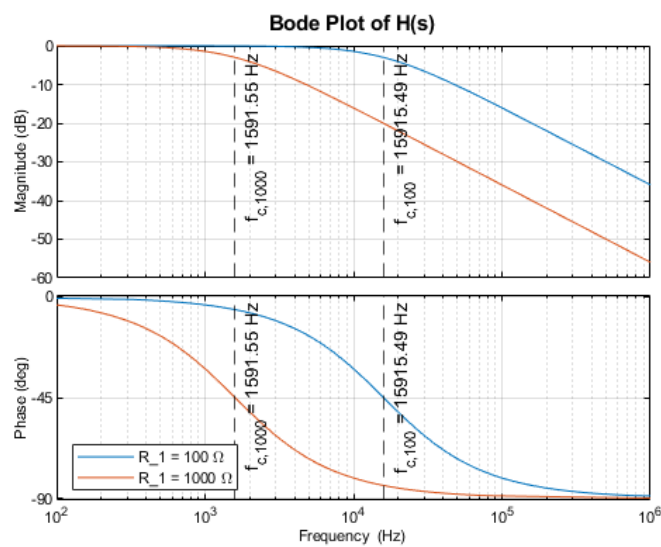


Figure 7

3. 3.1. Assign nodes.

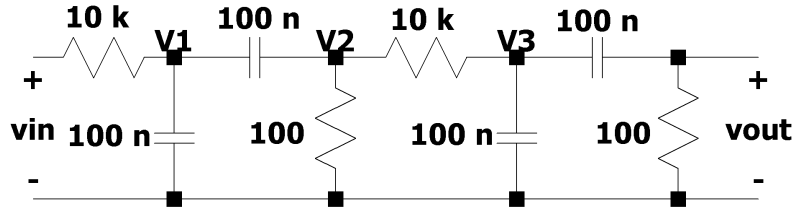


Figure 8

Let the impedance of each capacitor be Z_C in the s -domain. Apply KCL at each node.

$$\begin{aligned} \frac{V_1 - V_{in}}{10\text{ k}} + \frac{V_1}{Z_C} + \frac{V_1 - V_2}{100} &= 0, & \frac{V_2 - V_1}{Z_C} + \frac{V_2}{100} + \frac{V_2 - V_3}{10\text{ k}} &= 0, \\ \frac{V_3 - V_2}{10\text{ k}} + \frac{V_3}{Z_C} + \frac{V_3 - V_{out}}{100} &= 0, & \frac{V_{out} - V_3}{Z_C} + \frac{V_{out}}{100} &= 0. \end{aligned}$$

$Z_C = \frac{1}{sC} = \frac{1}{s \cdot 100 \cdot 10^{-9}} = \frac{10^7}{s}$. Equate and eliminate the denominators in each expression.

$$1000V_1 + s(V_1 + V_1 - V_2) = 1000V_{in}, \quad s(V_2 - V_1) + 100000V_2 + 1000V_2 - 1000V_3 = 0$$

$$1000V_3 - 1000V_2 + s(V_3 + V_3 - V_{out}) = 0, \quad s(V_{out} - V_3) + 100000V_{out} = 0.$$

Rearrange the equations and put in matrix form.

$$\begin{aligned} V_1(1000 + 2s) - V_2s &= 1000V_{in}, & V_1(-s) + V_2(s + 101000) - V_3 \cdot 1000 &= 0 \\ V_2(-1000) + V_3(1000 + 2s) - V_{out}s &= 0, & V_3(-s) + V_{out}(s + 100000) &= 0 \end{aligned}$$

$$\begin{bmatrix} 1000 + 2s & -s & 0 & 0 \\ -s & s + 101000 & -1000 & 0 \\ 0 & -1000 & 1000 + 2s & -s \\ 0 & 0 & -s & s + 100000 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_{out} \end{bmatrix} = \begin{bmatrix} 1000V_{in} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Code:

```
syms s Vin
A = [1000+2*s, -s, 0, 0;
     -s, s+101000, -1000, 0;
     0, -1000, 1000+2*s, -s;
     0, 0, -s, s+100000];
B = [1000*Vin; 0; 0; 0];
V = A\B;
Vout = V(4);
H_sym = simplify(Vout / Vin);
[num_sym, den_sym] = numden(H_sym);
num_vec = double(coeffs(num_sym, s, 'All'));
den_vec = double(coeffs(den_sym, s, 'All'));
H = tf(num_vec, den_vec)
```

```

H =

          1e06 s^2
-----
s^4 + 404000 s^3 + 4.1e10 s^2 + 4.04e13 s + 1e16

Continuous-time transfer function.
Model Properties
>> |

```

Figure 9

$$H(j\omega) = \frac{-10^6 \omega^2}{\omega^4 - 404000j\omega^3 - 4.1 \cdot 10^{10} \omega^2 + 4.04 \cdot 10^{13} j\omega + 10^{16}}$$

The poles are

$$p_1 = -2.0292 \cdot 10^5, \quad p_2 = -2.0009 \cdot 10^5, \quad p_3 = -0.0050 \cdot 10^5, \quad p_4 = -0.0049 \cdot 10^5.$$

The frequencies are then

$$f_1 = 3.2296 \cdot 10^4, \quad f_2 = 3.1845 \cdot 10^4, \quad f_3 = 0.0080 \cdot 10^4, \quad f_4 = 0.0078 \cdot 10^4$$

For the bandpass, f_2 and f_3 are meaningful. Therefore,

$$f_{c, \text{lower}} = 80 \text{ Hz}, \quad f_{c, \text{upper}} = 31845 \text{ Hz}.$$

The bandwidth:

$$f_{c, \text{upper}} - f_{c, \text{lower}} = 31845 - 80 = 31765 \text{ Hz}$$

This circuit is a bandpass filter that passes the specific range of frequencies and attenuates the other frequencies.

3.2. Code:

```

p = pole(H);
f_poles = abs(p) / (2*pi)

```

```

f_poles =

    1.0e+04 *

    3.2296
    3.1845
    0.0080
    0.0078

```

Figure 10

Since this is a fourth-order bandpass filter, we take the frequencies that are closer to the band.

$$f_{\text{cutoff, lower}} = 80 \text{ Hz}, \quad f_{\text{cutoff, upper}} = 31845 \text{ Hz}$$

3.3. Code:

```
scales = {'Log', 'Linear'};
figure;
for i = 1:2
    subplot(2, 1, i);
    bode_opt = bodeoptions;
    bode_opt.Grid = 'on';
    bode_opt.FreqScale = scales{i};
    bode_opt.FreqUnits = 'Hz';
    bode_opt.Title.String = ['Bode Plot (', scales{i}, ' Scale)'];
    bodeplot(H, bode_opt);
end

axes = findall(gcf,'type','axes');
for k = 1:length(axes)
    xline(axes(k),f_poles(3),'--k', ...
        sprintf('f_{c, upper} = %.2f Hz',f_poles(3)));
    xline(axes(k),f_poles(2),'--k', ...
        sprintf('f_{c, lower} = %.2f Hz',f_poles(2)));
end
```

The plots are given in the next page.

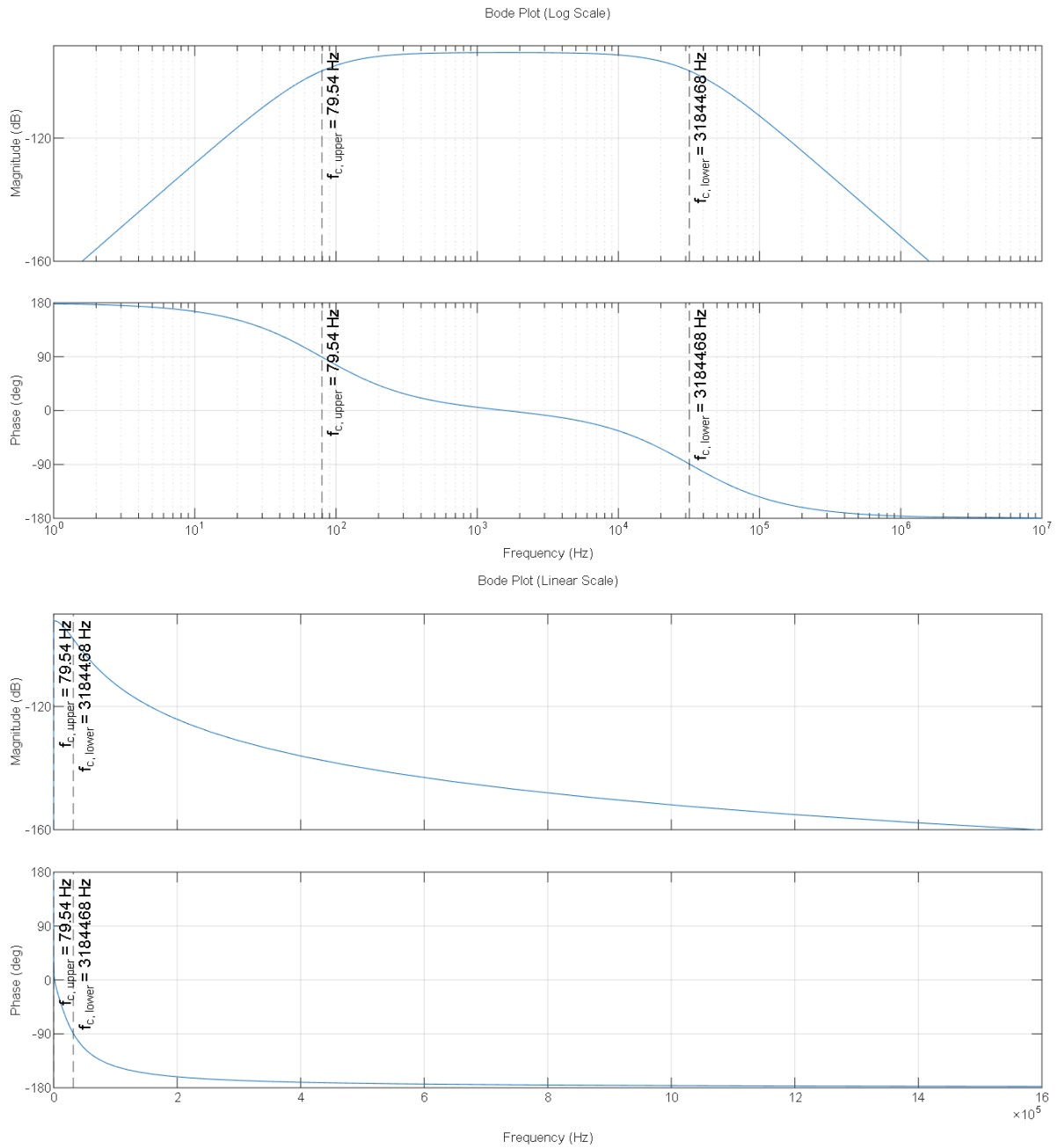


Figure 11

The main reason we use logarithmic scales in Bode plots is that they transform the complex exponential relationships of a circuit into straight lines, facilitating to visualize a filter's behavior across a range of frequencies. If we used the linear scale, we would struggle to analyze the cutoff frequencies as above.

The advantage of using high-order filters is that they select the certain frequency range that will be attenuated or passed, performing better compared to low-order ones. The disadvantage of using these filters is that the complexity of the circuit becomes higher, making calculations more difficult.